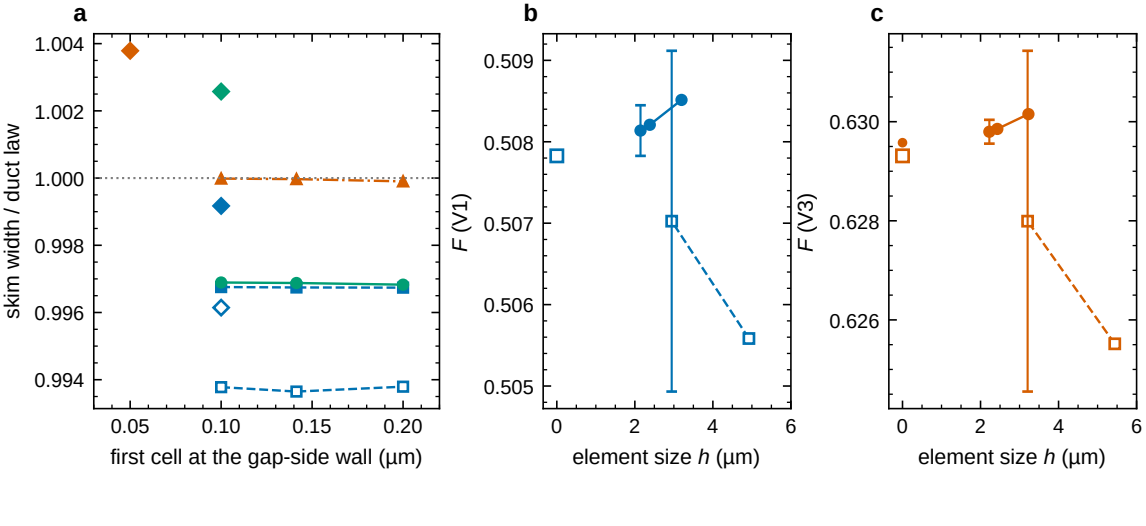
a
b
c
skim width / duct law
1.004
1.002
1.000
0.998
0.996
0.994
0.05
0.10
0.15
0.20
first cell at the gap-side wall (μm)
F (V1)
0.509
0.508
0.507
0.506
0.505
0
2
4
6
element size h (μm)
F (V3)
0.630
0.628
0.626
0
2
4
6
element size h (μm)
(a) run B, Python CFD
(a) run A, Python CFD
(a) run C, Python CFD
(a) open: run B, uniform inlet
(a) COMSOL, one mesh
(a) exact duct law
(b, c) Python CFD, three meshes
(b, c) COMSOL, two meshes
(b, c) Python CFD, extrapolated
(b, c) COMSOL, extrapolated
(b, c) ±u_num, finest mesh